Building a report

One document holding the report's supported figures, their parameters and its provenance, redrawn from the data as it stands when the document is built.

AT A GLANCE

Where	Summary → Report
Format	One self-contained HTML file. Static figures are embedded as images inside it; interactive ones as working plots. There is no PDF or Word output.
What it holds	A sample mapping, then one numbered block per loaded level: its fixed set of figures, three summary tables at protein and phosphosite level, and any logged volcano comparisons.
Chosen by the user	The title, the author, the text blocks, and which volcano comparisons are included. Nothing else is selectable.
Figures	Redrawn at build time, never stored. See *Captured or redrawn.*
In viewer mode	This tab is not shown.

1 What a report is

The report brings together a fixed selection of figures from the data that is loaded, each under a heading, each with the prose that belongs to it, in one file that can be sent to somebody who does not have the tool. It is not a figure collection assembled by hand. With one exception — the volcano — there is no list of items to add: a level that is loaded and annotated contributes its whole block, and a level that is not contributes nothing.

1. Upload the datasets and generate their annotations. This alone determines which blocks the report has.
2. Leave every page whose figure will be in it set the way it should appear. The report reads the last state of each page.
3. On Statistics → Volcano Plot, set up each comparison and press **Add Volcano To Report**. Comparisons not added do not appear.

4. On Summary → Text, write the introduction and any text that belongs above or below a figure.
5. Here, fill in **Report Title** and **Author**, press **Generate HTML Report**, and read the preview below.
6. Press **Download HTML Report** to save it.

2 Controls

Controls on this page

Control	What it does	Default	Where
Report Title	The heading of the document, its browser tab title, and the name of the downloaded file: everything that is not a letter, digit, hyphen or underscore becomes an underscore. Left blank, the document is titled Untitled Report.	empty	page
Author	Printed under the title. Left blank, it reads Unknown Author.	empty	page
Generate HTML Report	Builds the document. Every figure in it is drawn at this moment. Reads Generating… while it runs.	action	page
Download HTML Report	Saves the document that is currently in the preview. Appears only after a successful build, and saves that build — changing a setting afterwards does not change the file until the report is generated again.	action	page

On the volcano page

Control	What it does	Default	Where
Add Volcano To Report	Adds the comparison as it is currently configured. Refuses while the two conditions are not both chosen and different.	action	page
Remove Volcano From Report	Removes every logged comparison that matches the one currently configured.	action	page

The line beside the two buttons reads Saved entries for <level> and counts the comparisons

logged at the level currently selected, not in the whole session. Both buttons are hidden while **Source** is set to Volcano Control.

> NOTE
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> The report has no appearance controls of its own. Figure sizes, resolutions, colours and titles all come from the pages that produce the figures, and from the plot appearance panel described in Part XI.

3 Captured or redrawn

> THE RULE
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> No figure is ever stored. Every figure in the report is drawn when **Generate HTML Report** is pressed, from the data as it stands at that moment, with the settings as they stand at that moment. The only thing an item carries from the time it was added is the set of parameters listed below, and that applies to the volcano alone.

What follows from the rule, stated the other way round:

- Change a control on the coverage page, or a colour in the plot appearance panel, then build: the report's figures change, the volcano added a week ago included.
- Re-run the filter, change the imputation, re-annotate, or upload a different file: every number in the report changes, because every figure and every count is computed again from the current matrices. A logged volcano keeps its thresholds but not its results.
- Change a volcano threshold and build without pressing **Add Volcano To Report** again: the report is unaffected. The thresholds it uses are the ones stored with the entry.
- Turning advanced mode off hides its controls; stored appearance overrides still apply to the report. Reset an override to remove it.

> EASY TO GET WRONG
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> Because the settings are read live, only the *last* state of a page reaches the report. Two coverage plots of the same level with different options cannot both be in one report, and neither can two differently scaled PCAs. The volcano is the exception: each logged comparison carries its own parameters, and several can sit side by side in one section.

What a volcano entry stores

Column	What it holds
Level	Protein or phosphosite. The block the comparison is filed under.
Data Source	Data, Filtered Data or Imputed Data. An unrecognised stored value is read as Data rather than as the most processed matrix.
Condition 1, Condition 2	The comparison, in that order: $\log_2$ FC is condition 1 minus condition 2, so a feature reported as upregulated is higher in condition 1.
Label column	Which identifier labels the points.
Test type	Unpaired pooled, unpaired Welch, or paired.
P-value threshold, log2 FC threshold	As set, and whether **Use uncorrected p-values** was on — that is, whether the first threshold applies to the raw or to the Benjamini–Hochberg adjusted value.
Highlight terms	The list, whitespace-trimmed, empties dropped.
Type	Normal or Interactive — per comparison, so one report can hold a print figure and a hoverable one.
Label top up-regulated, Label top down-regulated	Stored only for a Normal entry; an Interactive entry stores zero for both, because only the static renderer draws labels.

Adding a comparison that matches one already logged in every one of those fields does nothing and says so. Removing works the same way round: it removes every entry matching the current configuration and reports how many. There is no list of logged comparisons on screen and no way to remove one except by reconstructing its settings, which is why the counter beside the buttons is worth watching.

> EASY TO GET WRONG
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> The log lives with the browser session, and it is cleared automatically whenever the loaded datasets change: a new upload, a different data package, a different metadata profile, or a file whose row or column count differs. That is deliberate — a comparison logged against one dataset would be redrawn against another — but it means the log is empty again after a re-upload, without announcement. It is written into a saved state file and comes back with it.

4 What the document contains

In order, from the top:

- The title, the author, the date the report was built, and a version line. It is the version of the application that built the report, which is the version on this manual's title page.
- Introduction — only if a text block was written for it.

- Table of Contents, listing the blocks and their numbered sections. A volcano section is listed only when that level has a logged comparison.
- Sample Mapping — one table per level giving, for every sample, its plot label condition_n, its original name in the uploaded file, its condition, and its pair when the annotation has pairing. Two levels with identical mappings share one table. This is the key to every axis label in the document.
- The level blocks, numbered from 1 in the order peptide, protein, phosphosite, with only the loaded ones present.

The document has no warnings section. Warnings are shown in the application only, in the **Report Warnings** panel on this page, and never written into the report.

Each level block opens with its filtering, normalization and imputation paragraphs, then gives its sections in a fixed order. A section is a heading, the fixed commentary, the Above text block, the figure, and the Below text block; the protein and phosphosite coverage, boxplot and coefficient-of-variation sections carry their summary table under the figure, and the PCA and volcano sections carry their provenance and statistics lines between the text block and the figure. Those sentences cite the published methods they describe — the imputation, normalization, test and correction, and the Class I localization threshold — in the text. All of that prose is described in *The methods text*.

> **WORTH KNOWING**
>
> A figure or table that cannot be drawn does not stop the build: a red box saying that it could not be rendered takes its place in the document, and the **Report Warnings** panel on this page names it as <part> could not be rendered: <reason>. The reason appears only in that panel, never in the report, so read the panel before sending the file.

The file

Beside **Download HTML Report**, the notebook download provides one ZIP for the plots and tables included in that report. It has numbered calculation sections, required helpers and resources, and input-file placeholders. Where a section uses imputed data, the exact matrix is supplied. Where it does not, the recipe starts from the original input and reproduces the required preprocessing. It does not add analyses that the report omitted.

The archive belongs to the built report snapshot, not to a later state of the session. If the application cannot construct a complete recipe, it reports that separately; an HTML report may still be available. Check both the **Report Warnings** panel and the notebook availability before handing over a reproducible analysis.

The document is a single HTML file with nothing beside it: static figures are images encoded into the file, interactive ones are complete plots in sandboxed frames that keep their hover, zoom and legend behaviour offline, and a content policy in the file forbids it from fetching anything over a network. It can therefore be opened on a machine that has never seen the tool; browser and print layout can still differ. The file can be large, because every figure is inside it. To obtain a PDF, open it in a browser and print it; an interactive figure prints as

whatever it currently shows, so set it first, or log that comparison as Normal instead.

5 What a report cannot include

- *Any level but the three.* There is no phosphoprotein block and no precursor block. A volcano logged at phosphoprotein level is stored and counted, and then never appears.
- *Control-corrected volcanoes.* The two buttons are hidden while **Source** is Volcano Control, and a control-corrected entry is skipped at build time.
- *Any statistical output but the volcano.* No ANOVA, no over-representation analysis, no pathway heatmap, no replicate and variance projection (a what-if projection, not a power calculation).
- *Whole sections of the tool.* No Venn diagram or Pearson comparison; no single-protein figure; no phosphosite distribution, phosphoprotein regulation or Kinase Library; no missing-value heatmap; no distribution, imputation or identifier translation output.
- *Tables other than its own.* It builds the sample mapping and, for the protein and phosphosite blocks, a coverage, boxplot and coefficient-of-variation summary. Nothing else, and neither table of *The summary tables*.
- *A second version of any figure, or a chosen subset.* One report holds at most one of each — the volcano excepted — and a figure of a loaded level cannot be left out.

Edge cases

- A dataset loaded without an annotation contributes no block, and the **Report Warnings** panel on this page says so rather than the build failing; the document itself does not mention it.
- The peptide block uses the protein annotation for its sample mapping whenever a protein dataset is loaded, and its own peptide metadata only in a peptide-only session. Its coverage plot is always drawn at peptide level; the precursor variant has no place in the document.
- The missing-value plot is styled once for all three levels: the completeness page holds one set of options, and all three use it.
- A page never visited in this session contributes its figure with the built-in defaults, not with whatever it showed last time the tool was opened.

Behind this page

Function	What it produces
report_function	Builds the whole document and returns it with its file name and, separately, the warnings shown on this page
addVolcanoReportEntry	The two buttons: the duplicate test and the stored entry
removeVolcanoReportEntry	
getSummaryReportContext	Everything read from the session at build time: page settings, logged comparisons, appearance settings
syncSummaryReportState WithDatasets	Clears the log when the loaded datasets change
run_volcano	Re-runs each logged comparison for its counts
volcano_png / volcano_html	The two renderers, chosen by the entry's Type
sample_label_frame	The sample mapping tables